

Rational Polytopes

$$P = \{Ax \leq b\}, \quad A, b \text{ have}$$

rational coefficients: $A_{ij}, b_i \in \mathbb{Q}.$

Punchline in Advance

If we have upper bounds on the
encoding lengths of individual facets or
vertices of a polyhedron, then the solvability
of weak problems implies the solvability
of strong problems.

[And can drop assumptions on full dimensionality
; bddness].

$$\langle A \rangle, \langle b \rangle \quad \text{vs} \quad \max_i \langle a_i \rangle, \langle b_i \rangle$$

Derived Convex Sets

$$K_1, K_2$$

$$K = K_1 + K_2$$

$$K = K_1 \cap K_2$$

$$K = \text{conv}(K_1 \cup K_2)$$

other operations.

Facet and Vertex Complexity — 1/2

Def'n : $P \subseteq \mathbb{R}^n$ a polyhedron, $\phi, v \in \mathbb{Z}_{\geq 0}$

(a) P has facet-complexity ϕ if $\exists$

system of inequalities in $\mathbb{Q}$, describing P

s.t. the encoding length of each inequality is $\leq \phi$.

(b) $P = \text{conv}\{V\} + \text{cone}\{E\}$ s.t. each vertex v and ray $r \in E$ has enc. length at most v : Then P has vertex-complexity v .

$$\left[\text{If } P = \mathbb{R}^n, \text{ require } \phi \geq n+1 \right] \wedge \left[\text{If } P = \emptyset, v \geq n \right]$$

Facet and Vertex Complexity — 2/2

Def'n: A well-described polyhedron P is a triplet (P, n, ϕ) where P has facet complexity $\leq \phi$.

• Lemma: If P has facet-complexity ϕ , then it has vertex complexity at most $\underline{4n^2\phi}$.
If P has vertex complexity v , then $\underline{\phi \leq 3n^2v}$.

• Lemma: If there is a description of P w/ facet complexity ϕ , then a standard rep'n, has facet complexity at most $\underline{35n^2 \cdot \phi}$.

Connecting to Weak/Strong Problems — 1/2

Lemma : If $P \subseteq \mathbb{R}^n$ is full dim'd
with vertex-complexity v , then all vertices
of $P \subseteq S(0, 2^v)$.

Lemma : For P as above, with facet compl $\leq \phi$
then P contains ball with radius $\geq 2^{-7n^3\phi}$.

Connecting to Weak/Strong Problems — 2/2

Lemma: $P \subseteq \mathbb{R}^n$ fact-compl ϕ

Let $y \in \mathbb{Q}^n$, s.t. entries of y have common denominator at most q .

$$\underline{\text{If}} \quad y \in S(P, \underline{\underline{\frac{1}{q} 2^{-2\phi}}}) \Rightarrow y \in P.$$

Lemma: $P \subseteq \mathbb{R}^n$, vertex-complexity ω ,

If $\vec{a}^T x \leq b$ is an integral inequality that is almost valid for P , i.e., $\vec{a}^T x \leq b + 2^{-(\omega+1)}$ holds $\forall x \in P$

$$\Rightarrow \vec{a}^T x \leq b, \quad \forall x \in P.$$

Summary

For well-described polyhedra.

- ① The 4 strong problems are equivalent
- ② Weak oracles solve the strong problems.